MarketDB — SQL Assignment

Questions · Expected Output Screenshots · SQL Solutions

- Database Schema
- Q1 — Orders with ≥ 1/3 of all products
- Q2 — Profit Histogram (bin width = 2000)
- Q3 — Customer of the Month (CTE + RANK)
- Q4 — Rank sub-categories by total profit
- Q5 — Top 20 orders by profit (DENSE_RANK + LIMIT)
- Q6 — Skyline query on customers (CTE + NOT EXISTS)
- Q7 — Sub-categories covering ~25% of profits (CUME_DIST)
- Q8 — Customers with no orders (LEFT JOIN)

Database Schema

Table Structure:

TABLE_NAME	COLUMN_NAME	DATA_TYPE	IS_NULLABLE
dim_cust	customer_name	text	YES
dim_cust	city	text	YES
dim_cust	state	text	YES
dim_cust	customer_segment	text	YES
dim_cust	cust_id	text	YES
dim_order	ord_id	text	YES
dim_order	order_date	datetime	YES
dim_order	order_priority	text	YES
dim_prod	prod_id	text	YES
dim_prod	product_category	text	YES
dim_prod	product_sub_category	text	YES
dim_shipping	ship_mode	text	YES
dim_shipping	ship_date	datetime	YES
dim_shipping	ship_id	text	YES
fact_sales	ord_id	text	YES
fact_sales	prod_id	text	YES
fact_sales	ship_id	text	YES
fact_sales	cust_id	text	YES
fact_sales	sales	double	YES
fact_sales	discount	double	YES
fact_sales	order_quantity	int	YES
fact_sales	profit	double	YES
fact_sales	shipping_Cost	double	YES
fact_sales	product_base_margin	double	YES

Question 1 — Orders with ≥ 1/3 of All Available Products

Expected Output (Screenshot):

Question **1**

Not yet answered

Marked out of 5.00

Flag question

Display the order IDs of orders for which the number of products in the order is at least one-third of the number of all available products.

```
+----------+
| ord_id   |
+----------+
| Ord_542  |
| Ord_2506 |
+----------+
2 rows in set (0.01 sec)
```

SQL Query:

```
SELECT ord_id
FROM fact_sales
GROUP BY ord_id
HAVING COUNT(prod_id) >= (SELECT COUNT(*) / 3 FROM dim_prod);
```

Question 2 — Profit Histogram (Bin Width = 2000)

Expected Output (Screenshot):

Question **2**

Not yet answered

Marked out of 5.00

Flag question

Generate a histogram for profit with a bin width of 2000 (for fact_sales table).

```
+-----------+-----------+
| bin-range | frequency |
+-----------+-----------+
|    -16000 |         1 |
|    -14000 |         1 |
|    -12000 |         5 |
|    -10000 |         4 |
|     -8000 |         4 |
|     -6000 |        10 |
|     -4000 |        37 |
|     -2000 |      4182 |
|         0 |      3796 |
|      2000 |       171 |
|      4000 |        57 |
|      6000 |        35 |
|      8000 |        22 |
|     10000 |         6 |
|     12000 |         3 |
|     14000 |         1 |
|     26000 |         1 |
+-----------+-----------+
17 rows in set (0.01 sec)
```

SQL Query:

```
SELECT
    FLOOR(profit / 2000) * 2000 AS `bin-range`,
    COUNT(*) AS frequency
FROM fact_sales
GROUP BY `bin-range`
ORDER BY `bin-range`;
```

Question 3 — Customer of the Month (Top 2 per Month)

Expected Output (Screenshot):

Question **3**

Not yet answered

Marked out of 5.00

Flag question

"Customer of the month" report

Based on the total sales done by each customer every month, create a report of TOP-2 (i.e., rank 1 and 2) customers for each month.

```
-------------+-----------+--------------------+-----------+
| sales_month | cust_id   | total_sales        | cust_rank |
+-------------+-----------+--------------------+-----------+
|           1 | Cust_942  | 38399.350000000006 |         1 |
|           1 | Cust_1797 |          21134.71 |         2 |
|           2 | Cust_1007 |         37406.646 |         1 |
|           2 | Cust_1746 |        24587.1855 |         2 |
|           3 | Cust_1799 |           29180.9 |         1 |
|           3 | Cust_1307 |          29005.65 |         2 |
|           4 | Cust_937  |         33902.363 |         1 |
|           4 | Cust_1812 |        30285.0705 |         2 |
|           5 | Cust_685  |          22950.87 |         1 |
|           5 | Cust_1464 | 20146.340000000004 |         2 |
|           6 | Cust_1310 |          29362.77 |         1 |
|           6 | Cust_92   |         26133.39 |         2 |
|           7 | Cust_932  |          48054.01 |         1 |
|           7 | Cust_63   |         23792.93 |         2 |
|           8 | Cust_1793 | 23823.485999999997 |         1 |
|           8 | Cust_1445 |          19023.48 |         2 |
|           9 | Cust_1571 |           29884.6 |         1 |
|           9 | Cust_62   |         22231.75 |         2 |
|          10 | Cust_247  |          27663.92 |         1 |
|          10 | Cust_1401 |          22045.92 |         2 |
|          11 | Cust_934  |         43138.169 |         1 |
|          11 | Cust_1421 | 41740.689999999995 |         2 |
|          12 | Cust_1151 |  96161.32400000001 |         1 |
|          12 | Cust_466  |         46682.2445 |         2 |
+-------------+-----------+--------------------+-----------+
24 rows in set (0.03 sec)
```

SQL Query:

```
WITH monthSelect AS (
    SELECT
        MONTH(o.order_date) AS sales_month,
        f.cust_id,
        SUM(f.sales) AS total_sales
    FROM fact_sales f
    JOIN dim_order o ON f.ord_id = o.ord_id
    GROUP BY MONTH(o.order_date), f.cust_id
),
rankCnt AS (
    SELECT *,
        RANK() OVER (
            PARTITION BY sales_month
            ORDER BY total_sales DESC
        ) AS cust_rank
    FROM monthSelect
)
SELECT sales_month, cust_id, total_sales, cust_rank
FROM rankCnt
WHERE cust_rank <= 2
ORDER BY sales_month, cust_rank;
```

Question 4 — Rank Product Sub-Categories by Total Profit

Expected Output (Screenshot):

Question **4**

Not yet answered

Marked out of 5.00

Flag question

Ranking the Product_Sub_Category based on the average profit made by each of them across all the sales i.e., ranks of the profitable product subcategory. Below is the result.

```
+-------------------------------+--------------------+-------------+
| product_sub_category          | profit             | profit_rank |
+-------------------------------+--------------------+-------------+
| TELEPHONES AND COMMUNICATION  | 316951.6200000003 |           1 |
| OFFICE MACHINES               | 307712.92999999993 |           2 |
| BINDERS AND BINDER ACCESSORIES | 307413.38999999996 |           3 |
| COPIERS AND FAX               | 167361.48999999996 |           4 |
| CHAIRS & CHAIRMATS            | 122738.0699999999 |           5 |
| OFFICE FURNISHINGS            | 100427.93 |           6 |
| APPLIANCES                    | 97158.05999999988 |           7 |
| COMPUTER PERIPHERALS          | 94287.48000000001 |           8 |
| ENVELOPES                     | 48182.60000000004 |           9 |
| PAPER                         | 45263.20000000003 |          10 |
| LABELS                        | 13677.16999999999 |          11 |
| STORAGE & ORGANIZATION        | 13599.489999999982 |          12 |
| PENS & ART SUPPLIES           | 7564.780000000003 |          13 |
| RUBBER BANDS                  | -102.6700000000001 |          14 |
| SCISSORS, RULERS AND TRIMMERS | -7799.250000000001 |          15 |
| BOOKCASES                     | -33729.09000000002 |          16 |
| TABLES                        | -113468.18000000001 |          17 |
+-------------------------------+--------------------+-------------+
17 rows in set (0.02 sec)
```

SQL Query:

```
SELECT
    p.product_sub_category,
    SUM(f.profit) AS profit,
    RANK() OVER (ORDER BY SUM(f.profit) DESC) AS profit_rank
FROM fact_sales f
JOIN dim_prod p ON f.prod_id = p.prod_id
GROUP BY p.product_sub_category
ORDER BY profit_rank;
```

Question 5 — Top 20 Orders by Dense Rank on Decreasing Profit

Expected Output (Screenshot):

Question **5**

Not yet answered

Marked out of 5.00

Flag question

Display the top 20 Sales and Discount made based on dense rank in the order of decreasing profit.

ord_id	sales	discount	profit	profit_rank
Ord_3084	89061.05	0	27220.69	1
Ord_3707	28359.4	0.05	14440.39	2
Ord_3143	28664.52	0.09	13340.26	3
Ord_4614	29884.6	0.05	12748.86	4
Ord_5186	26095.13	0.03	12606.81	5
Ord_5425	27720.98	0.07	11984.4	6
Ord_2425	27820.34	0.08	11630.15	7
Ord_3727	29186.49	0.05	11562.08	8
Ord_2345	25409.63	0.02	11535.28	9
Ord_118	23792.93	0	10951.31	10
Ord_3059	21921.28	0.04	10521.33	11
Ord_1265	24051.49	0.07	9791.04	12
Ord_2753	23106.46	0.08	9527.47	13
Ord_4216	26126.92	0.04	9498.6	14
Ord_3067	20175.48	0	9373.96	15
Ord_5361	21752.01	0.03	9296.35	16
Ord_1920	20329.8	0	9224.1	17
Ord_882	23281.05	0.09	9097.65	18
Ord_2872	21555.6	0.02	8965.83	19
Ord_1764	18697.24	0.02	8918.74	20

20 rows in set (0.01 sec)

SQL Query:

```
SELECT
    ord_id, sales, discount, profit,
    DENSE_RANK() OVER (ORDER BY profit DESC) AS profit_rank
FROM fact_sales
ORDER BY profit DESC
LIMIT 20;
```

Question 6 — Skyline Query on Customers

Expected Output (Screenshot):

Question **6**

Not yet answered

Marked out of 5.00

Flag question

You are the director of popular e-commerce company KlipDart. You want to target a set of customers (**skyline customers**) to encourage them to buy more often because your company makes a lot of profit and they order in bulk when they buy from you. A skyline customer is characterized by 3 parameters: Total Profit ("total_profit") made so far from the customer (Sum of profits made for a customer). Total Order_Quantity ("total_quantity") done so far by the customer (Sum of Order_Quantity made for the customer). Total number of orders ("total_orders") placed (count of orders). Perform a skyline query based on customers using the total_profit, total_quantity,, total_orders parameters and print the output in below format (Exact output expected). NOTE: A skyline query returns a set of points that are not dominated by any other point A point dominates another point if it is as good or better in ALL the dimensions

```
+-----------+--------------------+----------------+--------------+
| cust_id   | total_profit       | total_quantity | total_orders |
+-----------+--------------------+----------------+--------------+
| Cust_444  | 4708.150000000001 |          548 |         21 |
| Cust_1151 |          28663.71 |          129 |          6 |
| Cust_63   | 20877.440000000002 |          242 |          7 |
| Cust_1329 |           4533.58 |          554 |         21 |
| Cust_572  | 2774.8000000000006 |          586 |         21 |
| Cust_1421 |          18960.63 |          434 |         15 |
| Cust_1140 | 167.35999999999996 |          713 |         30 |
| Cust_1799 | 18760.589999999997 |          567 |         20 |
+-----------+--------------------+----------------+--------------+
8 rows in set (0.02 sec)
```

SQL Query:

```
WITH cust_metrics AS (
    SELECT
        cust_id,
        SUM(profit) AS total_profit,
        SUM(order_quantity) AS total_quantity,
        COUNT(ord_id) AS total_orders
    FROM fact_sales
    GROUP BY cust_id
)
SELECT *
FROM cust_metrics c1
WHERE NOT EXISTS (
    SELECT 1
    FROM cust_metrics c2
    WHERE
        c2.total_profit >= c1.total_profit
        AND c2.total_quantity >= c1.total_quantity
        AND c2.total_orders >= c1.total_orders
        AND (
                c2.total_profit > c1.total_profit
             OR c2.total_quantity > c1.total_quantity
             OR c2.total_orders > c1.total_orders
            )
        AND c2.cust_id != c1.cust_id
);
```

Question 7 — Sub-Categories Accounting for ~25% of Profits

Expected Output (Screenshot):

Question **7**

Not yet answered

Marked out of 5.00

Flag question

What product sub categories account for about 25% of the profits? (Hint: use cume_dist() and use "<= 0.25" as the cumulative distribution threshold)

```
+-------------------------------+---------------------+
| product_sub_category          | distribution        |
+-------------------------------+---------------------+
| TELEPHONES AND COMMUNICATION  | 0.058823529411764705 |
| OFFICE MACHINES               |  0.11764705882352941 |
| BINDERS AND BINDER ACCESSORIES |  0.17647058823529413 |
| COPIERS AND FAX               |  0.23529411764705882 |
+-------------------------------+---------------------+
4 rows in set (0.02 sec)
```

SQL Query:

```
WITH subcat_profit AS (
    SELECT
        p.product_sub_category,
        SUM(f.profit) AS cat_profit
    FROM fact_sales f
    JOIN dim_prod p ON f.prod_id = p.prod_id
    GROUP BY p.product_sub_category
),
ranked AS (
    SELECT
        product_sub_category,
        cat_profit,
        CUME_DIST() OVER (ORDER BY cat_profit DESC) AS distribution
    FROM subcat_profit
)
SELECT *
FROM ranked
WHERE distribution <= 0.25;
```

Question 8 — Customers Who Did Not Place Any Order

Expected Output (Screenshot):

Question **8**

Not yet answered

Marked out of 5.00

Flag question

Display details of customers who did not place even a single order

```
+-----------+----------------+----------+-------------+------------------+
| cust_id   | customer_name  | city     | state       | customer_segment |
+-----------+----------------+----------+-------------+------------------+
| Cust_6    | MONICA FEDERLE | Kolkatta | West Bengal | CORPORATE        |
| Cust_210  | BRIAN MOSS     | Chennai  | Tamil Nadu  | CORPORATE        |
| Cust_1584 | JONI WASSERMAN | Chennai  | Tamil Nadu  | CONSUMER         |
+-----------+----------------+----------+-------------+------------------+
3 rows in set (10.33 sec)
```

SQL Query:

```
SELECT
    c.cust_id,
    c.customer_name,
    c.city,
    c.state,
    c.customer_segment
FROM dim_cust c
LEFT JOIN fact_sales f ON c.cust_id = f.cust_id
WHERE f.cust_id IS NULL;
```